



THE UNITED REPUBLIC OF TANZANIA
MINISTRY OF
EDUCATION, SCIENCE AND TECHNOLOGY
**DAR ES SALAAM INSTITUTE OF
TECHNOLOGY**



DATA MINING AND ANALYTICS (MODULE COU 08104)
ASSIGNMENT 2 (BENG22COE & BENG22ETE)

18th December 2025

INSTRUCTIONS

- **This is an individual assignment.** Students are required to complete it independently.
- **Submissions** should be made via GitHub. I will compute a grade based on the state of your repo *as of the submission deadline*, so feel free to make changes to it up to that point.
- All answers to explanatory/analytical questions should be submitted in a single pdf file .
- Submit a PDF file and Python code file (. ipynb), [Do not Submit checkpoints for .ipynb] to **GitHub**
- Submit a single zip file containing all csv dataset files used and Python code file (.py) to **Gradescope**.
- Do not commit inoperative codes. You should document or comment on your codes professionally, this demonstrates that the code is unique and owned by the student, but avoid unnecessary comments, you should not write comments on obvious codes.
- **All import statements** must appear together in a single cell, or in consecutive cells at the top of the ipynb file, **before any other executable code**.
- Submitting answers directly from generative AI is not allowed as it could lead to being flagged for plagiarism and you will earn 0 score for the assignment.
- You should create a directory named **Assignment02** in your GitHub repository and your solution file should reside in this directory. The file **MUST** be named the same as your registration number, e.g., “230454545669.ipynb”.

The submission deadline is Thursday 11th, January 2026 11:59 PM EAT.

1. Diabetes data (30 points)

In this question we will be looking at data analyzing diabetes patients ($N = 442$). The data consists of 11 columns. These columns are AGE, SEX, BMI (body mass index), BP (average blood pressure), S1, S2, S3, S4, S5, S6 (the last six are blood serum measurements looking for example at the Glucose level in blood). The 11th column is the dependent variable y , a quantitative measure of disease progression 1 year after baseline (one year after incurring the disease).

1.1. Load *the diabetes data* into Python. Produce a correlation matrix of the explanatory variables. Make a heat-map of the matrix (using `imagesc` and `colorbar`) and describe the relationships between the variables.

1.2. What is collinearity? What effect does collinearity amongst predictor features/ variables have on their estimated coefficient value?

1.3. Create a multivariate linear model using all ten variables and a constant. What are the Mean Squared Error and the adjusted R^2 for your model? Are all variables significant? Could this be a problem of collinearity?

1.4. What is the difference between forward selection and backward selection?

1.5. How does the approach stepwise work in the sense of selecting variables? Use the stepwise function to interactively compose a model using forward selection. Which variables are selected? How does this function work? What is the MSE and R^2 value for this new model?

2. Analyzing the Titanic data set (35 points)

The `titanic3` data frame describes the survival status of individual passengers on the British passenger liner RMS Titanic when sunk. The data frame does not contain information for the crew, but it does contain actual and estimated ages for almost 80% of the passengers. More details about the data frame can be found on the file `titanic3info.txt`. For this question, use the `titanic` dataset stored in `titanic3.csv`.

2.1 What is the difference between logistic regression and linear regression?

2.2 Load in the `titanic` dataset and calculate the probability of survival for a passenger on the `titanic`.

2.3 Provide a table giving survival probabilities broken down by passenger class, gender and age.

2.4 Build a logistic regression model for survival rates based on passenger class, sex and age. What are the parameter estimates and are these parameters statistically significant?

2.5 What is the performance of the model, measured by classification accuracy (number of correct classifications divided by total number of classifications) based on confusion matrix?

3. PCA (35 points)

3.1 Give a qualitative description of Principal Component Analysis (PCA) and its applications in machine learning. Why might it be useful to consider PCA to transform a set of explanatory variables?

3.2 Write down the mathematical equations for PCA explaining how one transforms the raw input data matrix X into a new set of variables. Give an interpretation of each matrix.

3.3 Use Python **yfinance** to download the 30 Dow Jones constituent stocks ('MMM', 'AXP', 'AMGN', 'AAPL', 'BA', 'CAT', 'CVX', 'CSCO', 'KO', 'DIS', 'DOW', 'GS', 'HD', 'HON', 'IBM', 'INTC', 'JNJ', 'JPM', 'MCD', 'MRK', 'MSFT', 'NKE', 'PG', 'CRM', 'TRV', 'UNH', 'VZ', 'V', 'AMZN', 'WMT').

Use at least one year of daily returns (1st January 2020 to 1st January 2021) to calculate the correlation matrix for the 30 stocks that are constituents of the Dow Jones Index. Use this correlation matrix for PCA and construct bar graphs to show the weight of each stock for the first and second principal components. Is the first or second principal component similar to the market (equal weight on each stock)? Discuss why?

3.4 Calculate the amount of variance explained by each principal component and make a 'Scree' plot. How many principal components are required to explain 95% of the variance?

3.5 Investigate the scatter plot of the first two principal components and calculate the average of all 30 stocks. Based on Euclidean distances away from this average, identify the three most distant stocks. Can you explain why these stocks are unusual?